Electrical Power

Exam code: 0625 & 0972

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Electrical power equation

- The **power** of an appliance is defined as:

The rate at which energy is transferred by an appliance

- Power can be calculated in terms of energy:

$$P = \frac{E}{t}$$

- Where:
  - P = power, measured in watts (W)
  - E = energy transferred, measured in joules (J)
  - t = time, measured in seconds (s)
- Power can also be calculated in terms of work done:

$$P = \frac{W}{t}$$

- Where:
  - W = work done, which is equivalent to energy transferred, measured in joules (J)
- The power of an electrical device is the **energy transferred per second** by the device
  - The watt (W) is equivalent to joules per second (J/s)
- The power dissipated by an electrical component can be calculated by:

$$P = IV$$

- Where:
  - P = dissipated power, measured in watts (W)
  - I = current, measured in amps (A)
  - V = potential difference, measured in volts (V)

Worked Example

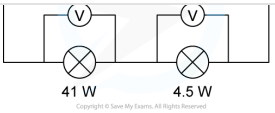
Two lamps are connected in series to a 150 V power supply.

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it



Which statement most accurately describes what happens?

A. Both lamps light normally

B. The 15 V lamp blows

C. Only the 41 W lamp lights

D. Both lamps light at less than their normal brightness

Answer: A

Step 1: State the equation for electrical power and rearrange for current

$$P = IV$$

$$I = \frac{P}{V}$$

Step 2: Calculate the current required for both lamps to operate

- For the 41 W lamp, with a voltmeter reading of 135 V:

$$I = \frac{41}{135} = 0.3 \text{ A}$$

- For the 4.5 W lamp, with a voltmeter reading of 15 V

$$I = \frac{4.5}{15} = 0.3 \text{ A}$$

Step 3: Determine the outcome of the bulbs

- For both bulbs to operate at their normal brightness, a current of 0.3 A is required
- The lamps are connected in series, so the same current would flow through both
- Therefore, the lamps will light at their normal brightness
  - This is option **A**

Examiner Tips and Tricks

When doing calculations involving electrical power, remember the unit is Watts W, therefore, you should **always** make sure that the time is in **seconds**

Measuring energy usage

Energy measured in joules

- Consider an average lightbulb with a power of 60 W, which is left on for 6 hours in a house
  - 1 hour is 3600 s
  - The energy transferred over this time is 1.296×10^6 J
- This number is large, and that is only one lightbulb for a single day
  - A household uses many appliances all year round; the energy transferred per month in joules would be inconveniently large

Energy measured in kilowatt-hours

- To make these large values more relatable to daily use:
  - Power can be measured in kilowatts (kW)
  - Time can be measured in hours (h)
- In this case, energy has units of kilowatt-hours (kW h)
  - The lightbulb from before receives 0.36 kW h of energy over the 6 hours
- This value is much easier to understand for consumers and energy providers; thinking in terms of hours of use is more practical than seconds

Calculations with kWh

- As has been stated previously, the equation for energy transferred is:

$$E = Pt$$

- But here, different units are considered:
  - E = energy transferred, measured in kilowatt hours (kW h)
  - P = power of the appliance, measured in kilowatts (kW)
  - t = time, measured in hours (h)
- The usual unit of energy is joules (J), which is one watt-second
- To find the number of joules in 1 kW h, convert the power and time to watts and seconds

$$1\ \text{kW h} = 1000\ \text{W} \times 3600\ \text{s} = 3.6 \times 10^6\ \text{J}$$

- 1000 watts multiplied by 3600 seconds is equal to 1000 multiplied by 3600, in watt-seconds

$$1000\ \text{W} \times 3600\ \text{s} = 1000 \times 3600\ \text{Ws} = 3.6 \times 10^6\ \text{J}$$

- Therefore, 1 kWh = 3.6×10^6 J
- To convert from kW h to J:

$$E\ (\text{kW h}) \times (3.6 \times 10^6) = E(\text{J})$$

- To convert from J to kW h:

$$E(\text{J}) \div (3.6 \times 10^6) = E(\text{kW h})$$

- The kW h is a large unit of energy, and is mostly used for energy in homes, businesses and factories

Worked Example

A cooker transfers 1.2×10^9 J of energy electrically to the thermal store of the heating element during its use over a year.

Assume that 1 kW h costs 14.2 p.

100 p = £1 (100 pence = 1 pound)

Calculate the cost of using the oven for the year.

Answer:

Step 1: List the known quantities

- Energy in joules, $E(\text{J}) = 1.2 \times 10^9 \text{ J}$
- Cost per kW h, $1 \text{ kW h} = 14.2 \text{ p}$

Step 2: Convert the energy used from J to kW h

$$E(\text{kW h}) = \frac{1.2 \times 10^9}{3.6 \times 10^6} = 333.3 \text{ kW h}$$

Step 3: Calculate the price

$$1 \text{ kW h} = 14.2 \text{ p}$$

$$333.3 \times 14.2 = 4733 \text{ p} = £47.33$$

Examiner Tips and Tricks

The kilowatt hour is a tricky concept to get your head around, so make sure you are comfortable with the conversions between kW h and J well before your exam.

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